

2020A F=ma Exam: Problem 22

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We have momentum conservation so $\Delta \vec{P} = 0$ in any inertial frame. We have

$$K_i = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

$$K_f = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2$$

so

$$\begin{aligned} \Delta K &= \frac{1}{2}m_1(v_{1f}^2 - v_1^2) + \frac{1}{2}m_2(v_{2f}^2 - v_2^2) = \frac{1}{2}m_1(\vec{v}_{1f} - \vec{v}_1) \cdot (\vec{v}_{1f} + \vec{v}_1) + \frac{1}{2}m_2(\vec{v}_{2f} - \vec{v}_2) \cdot (\vec{v}_{2f} + \vec{v}_2) \\ &= \frac{1}{2}\Delta \vec{p}_1 \cdot (\vec{v}_{1f} + \vec{v}_1) + \frac{1}{2}\Delta \vec{p}_2 \cdot (\vec{v}_{2f} + \vec{v}_2) = \frac{1}{2}\Delta \vec{p}_1 \cdot (\vec{v}_{1f} + \vec{v}_1 - \vec{v}_{2f} - \vec{v}_2) \end{aligned}$$

where we used the fact that $\Delta \vec{p}_1 = -\Delta \vec{p}_2$ by momentum conservation. We see that ΔK remains invariant under boosts of any velocity $\vec{u}$ i.e. adding $\vec{u}$ to all the velocity vectors.

Thus, $\Delta \vec{P}$ and ΔK are independent of inertial frame so the answer is A.